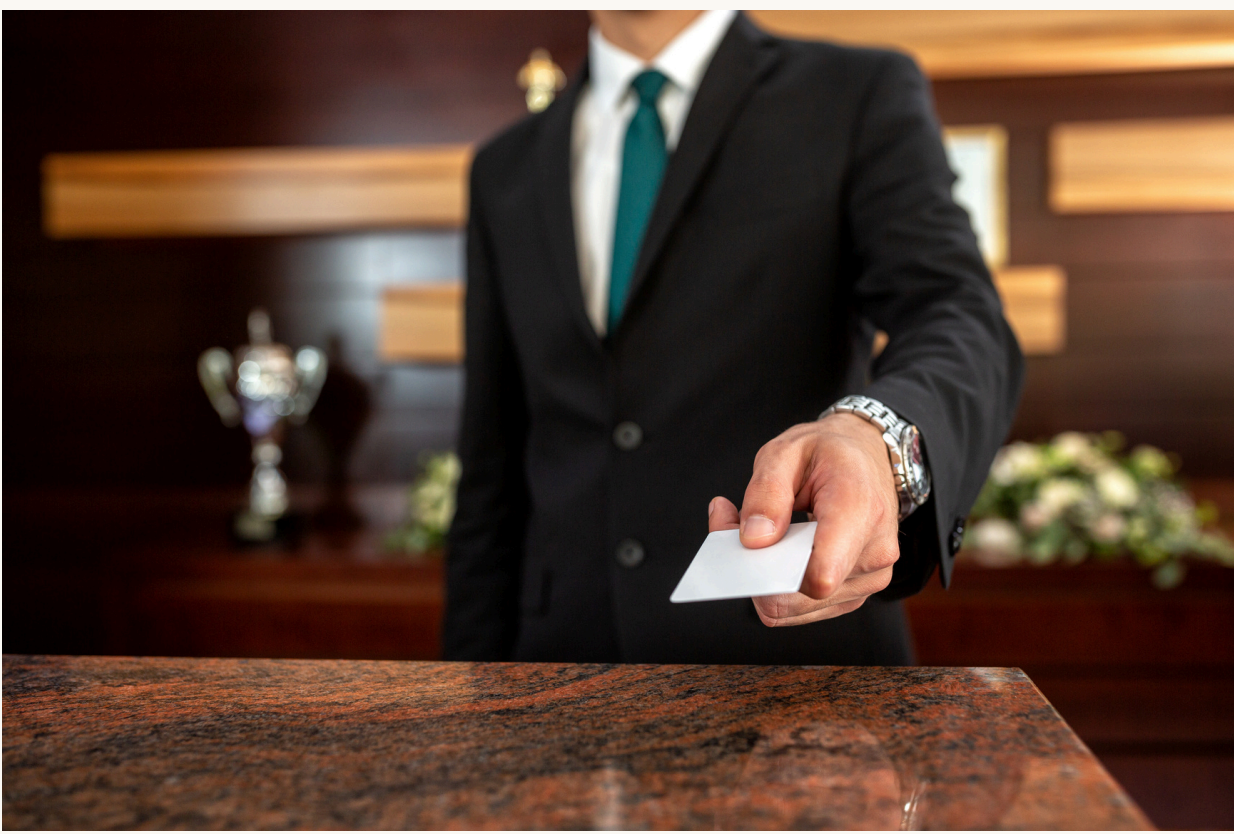


WHAT PREDICTS HOTEL BOOKING CANCELLATIONS?



GROUP $p < 0.05$
Alistair Knewstubb · Xiaotong Liu · Viktoriia Popova · Richa Purohit · Lavinia Roscia

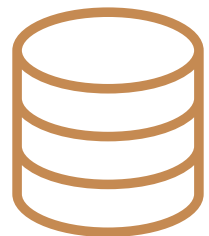


RESEARCH QUESTION

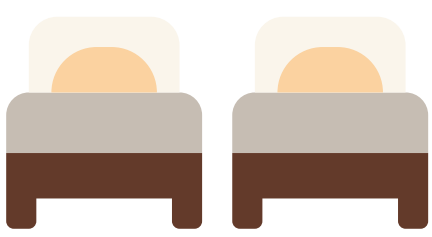
Which customer and booking characteristics are the strongest predictors of hotel booking cancellations?



DATA AT A GLANCE



119,390 bookings



2 hotel types
City + Resort



37.0% bookings cancelled

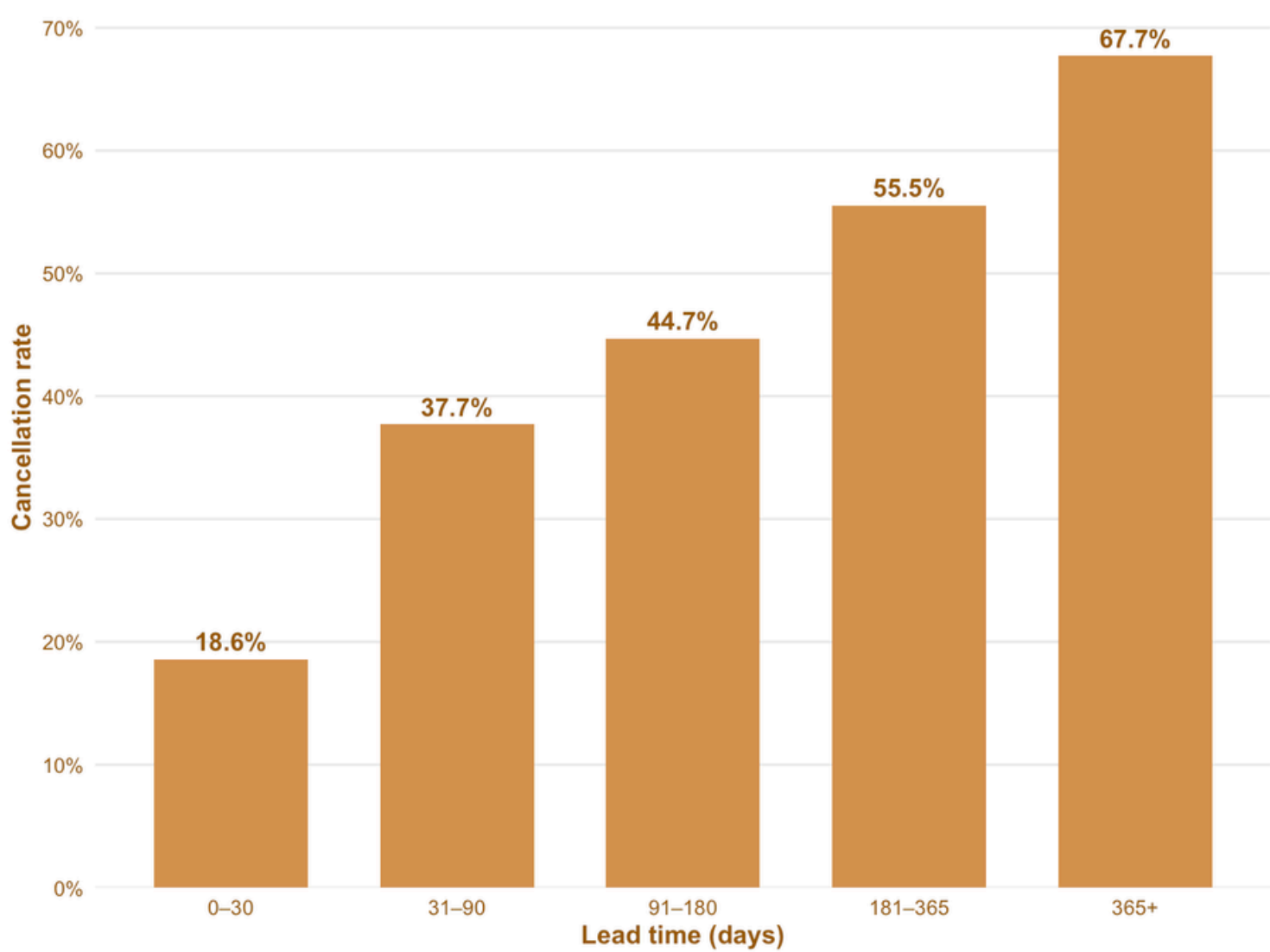
WHY IT MATTERS

Cancellations create uncertainty in hotel occupancy and make planning more difficult. Identifying higher-risk bookings could help hotels manage capacity and resources more effectively.

01 • THE FIRST CLUE

Longer lead time, higher cancellation

We first explored whether cancellation rates changed with how far in advance a booking was made.



Cancelled bookings had a median lead time of 113 days, compared with 45 days for non-cancelled bookings.

02 • THE SURPRISE

Non Refund bookings behaved very differently

We compared cancellation rates across deposit types and found a striking difference.

99.4% of Non Refund bookings were cancelled

14,494 of 14,587 bookings

Compared with **28.4% for No Deposit** and **22.2% for Refundable bookings**. In this dataset, Non Refund means a deposit equal to or greater than the total stay cost.

Why was the cancellation rate so extreme?

03 • DIGGING DEEPER

Could anything explain the 99.4%?

WE CHECKED

- **Market segment**

Groups and Offline TA/TO accounted for 97.2% of Non Refund bookings.

- **Previous cancellations:**

Even with no previous cancellation history, the rate barely changed:

99.4% → 99.2%

- **Time:**

Monthly cancellation rates remained **94.8–100%**

The extreme cancellation pattern persisted across all checks.

05 • CAN WE DO BETTER?

Comparing predictive performance

Finally, we compared logistic regression with tree-based methods to assess their predictive performance.

LOGISTIC

0.822

10-fold CV

TREE

0.819

10-fold CV

RANDOM FOREST

0.808

OOB

BOOSTING

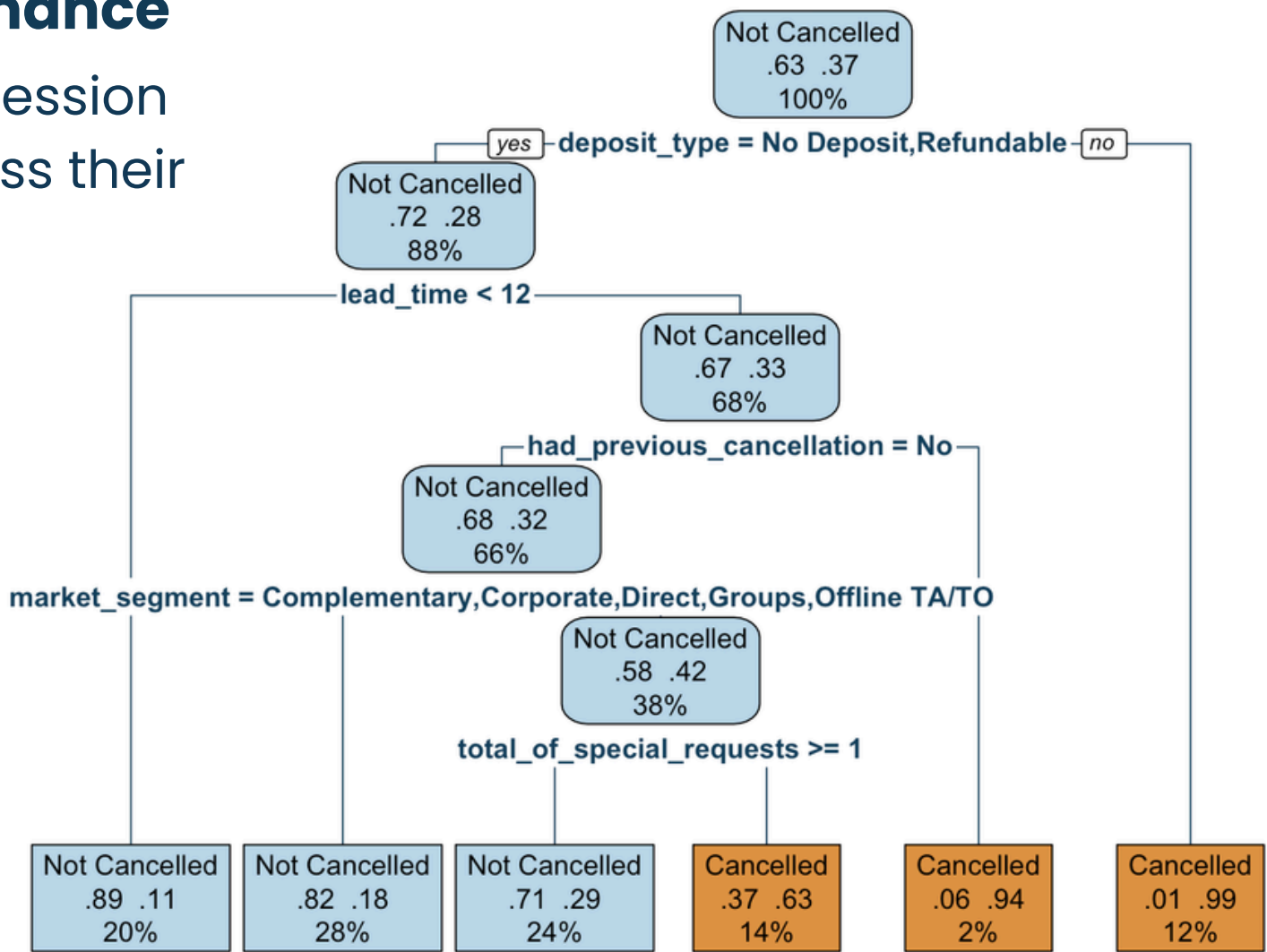
0.849

10-fold CV

The classification tree shows how deposit type, lead time, previous cancellations, market segment and special requests were used to split bookings.

Boosting achieved the highest discrimination (CV AUC = 0.849), while the simpler logistic regression remained competitive (CV AUC = 0.822).

HOW THE CLASSIFICATION TREE SPLIT THE DATA



LIMITATIONS

Data come from two Portuguese hotels (2015–2017), so results may not generalise to other hotels or markets. The available variables could not explain the extreme Non Refund cancellation pattern.

Data source: Antonio, Almeida & Nunes (2019), Hotel booking demand datasets, Data in Brief, 22, 41–49.

TAKEAWAY

Deposit type was the strongest predictive signal, while lead time, previous cancellation history, market segment and special requests also contributed to predicting cancellations.

These characteristics could help identify bookings at higher risk of cancellation and support hotel occupancy and booking planning. However, the extreme cancellation pattern among Non Refund bookings remains unexplained.